

TM Forum Technical Specification

Digital Identity Management

TMFC020

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Notice

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Table of Contents

Notice	2
Table of Contents	4
1. Overview	5
2. eTOM Processes, SID Data Entities and Functional Framework Functions....	6
2.1. eTOM business activities	6
2.2. SID ABEs.....	6
2.3. eTOM L2 - SID ABEs links.....	6
2.4. Functional Framework Functions	7
3. TM Forum Open APIs & Events	9
3.1. Exposed APIs	9
3.2. Dependent APIs.....	10
3.3. Events.....	11
4. Machine Readable Component Specification	12
5. References.....	13
5.1. TMF Standards related versions	13
5.2. Further resources.....	13
6. Administrative Appendix.....	14
6.1. Document History	14
6.1.1. Version History.....	14
6.1.2. Release History.....	14
6.2. Acknowledgements.....	15

1. Overview

Component Name	ID	Description	ODA Function Block
Digital Identity Management	TMFC020	The Digital Identity Management is responsible for the parties (customers, employees, partners) and resources authentication.	Party Management

TMFC020: Digital Identity Management



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

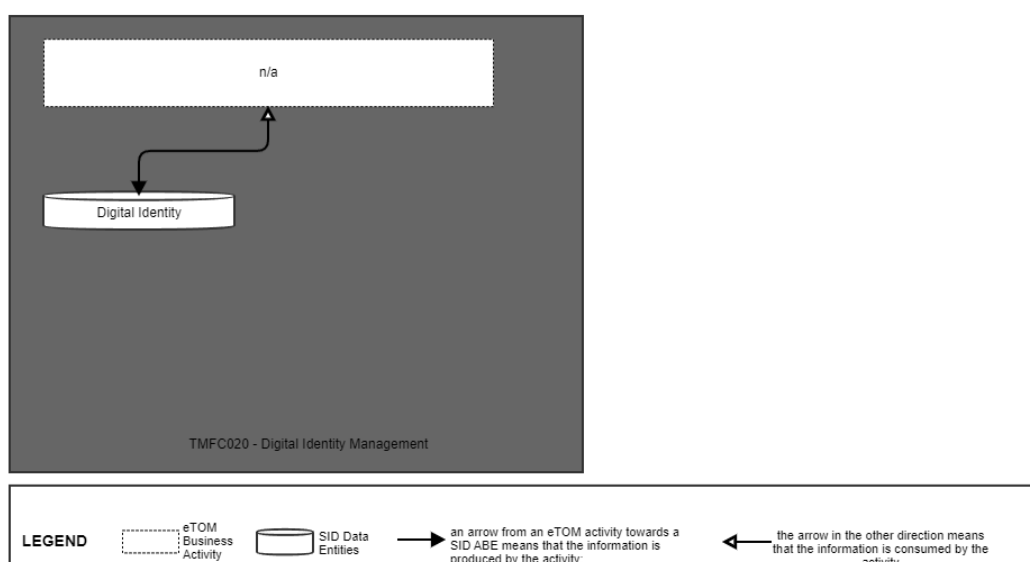
Identifier	Level	Business Activity Name	Description
n/a	2	n/a	Note: create JIRA issue to eTOM team to have process for Digital Identity Management

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)
Digital Identity	n/a

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Sub-Domain Functions Level 1	Sub-Domain Functions Level 2
1025	Application Access	Application Access provide access interfaces with authentication and authorization control for the requests and responses related to the application's functionality, including event logging and usage statistics.	Identification and Permission Management Identification and Authentication	Identification and Authentication Identification and Permissions
906	PKI and Digital Certificates Systems Integration	PKI and Digital Certificates Systems Integration provides integration to Public Key Infrastructure Systems that provides digital certificates, and the support to use public keys and digital certificates.	Identification and Permission Management Identification and Authentication	Identification and Authentication Identification and Permissions
899	Single Sign-On Access Control	Single Sign-On Access Control grant access in cooperation with central Authentication and Authorization functions to secure the most updated security.	Identification and Permission Management Identification and Authentication	Identification and Authentication Identification and Permissions
897	Building Access Control	Building Access Control checks, stops or allow physical access to facilities according to access roles and rules.	Identification and Authentication	Permission Control
898	Application Security Management	Application Security Management administrates the roles and rules that applies to getting the right to use an application.	Identification and Authentication	Permission Definition
1240	Identity Verification	Identity Verification (aka authentication) establishes that an actor (i.e., a person or a resource) is who they purport to be using one or several credentials. According to the reliability of the credentials used the authentication has a level of authentication such as a biometric credential (i.e., facial recognition or	Identification and Permission Management	Identification and Authentication

Function ID	Function Name	Function Description	Sub-Domain Functions Level 1	Sub-Domain Functions Level 2
		fingerprint) or multi-factor credential with a high level of authentication or a network credential with a low level of authentication. Note: each permission specifies the minimum level of authentication required.		
1247	Credentials Establishment	Credentials Establishment creates and/or modifies credentials and associates them with the Digital Identity that will be using them. Credentials can include username/passcode combinations, biometrics, and physical and/or logical passkeys. Note: A Digital Identity aims to enable identifying Party, Party Roles or Resource Roles.	Identification and Permission Management	Digital Identity Management
1248	Credentials Query	Credentials Query provides the ability to retrieve non-protected information about the Digital Identity of credentials or about the credentials themselves (e.g.name, photo, badge number, token ID, credentials valid dates, etc.).	Identification and Permission Management	Digital Identity Management

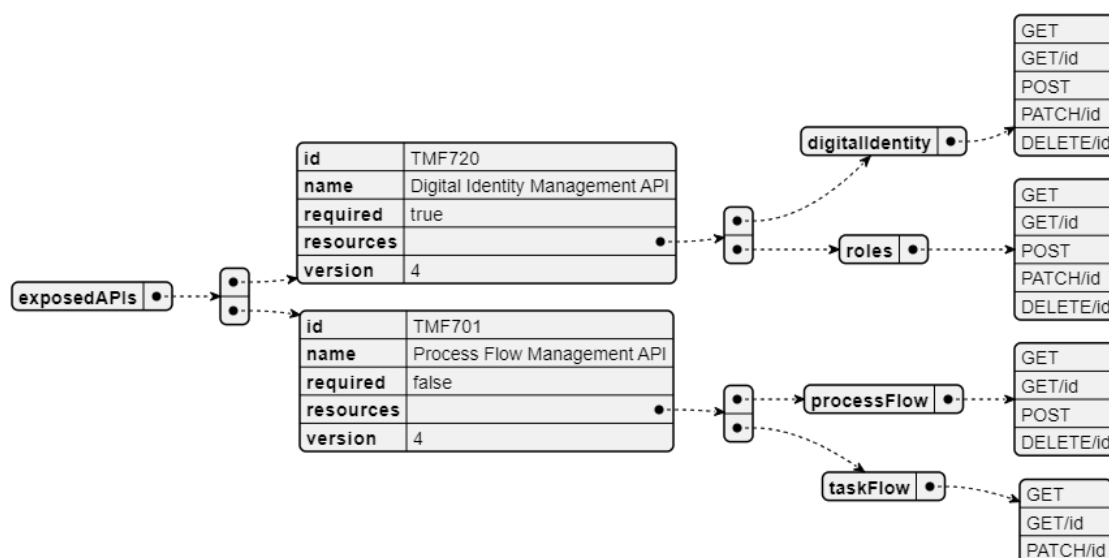
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

The following diagram illustrates API/Resource/Operation:

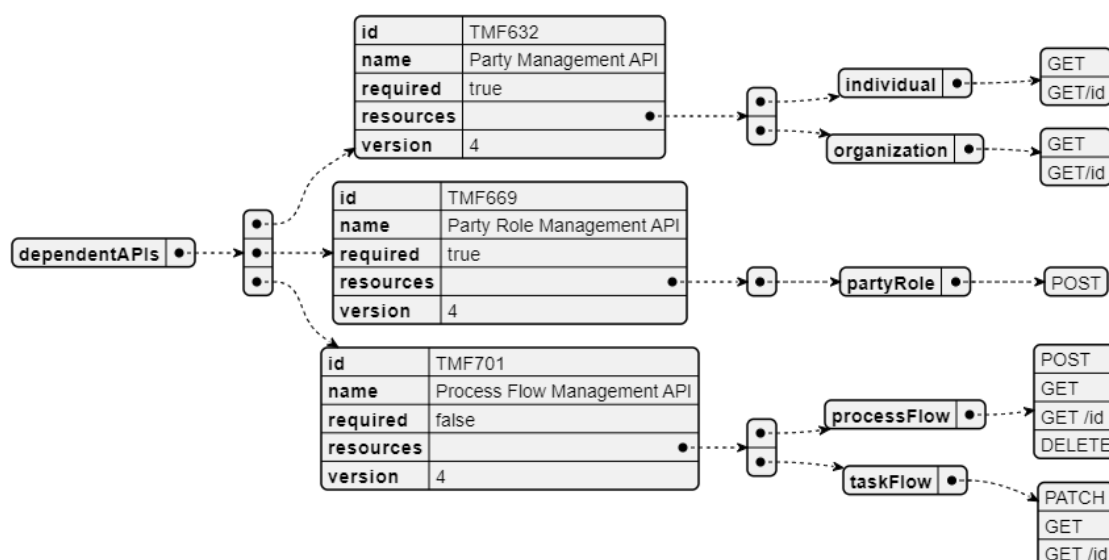


API ID	API Name	Mandatory / Optional	Operations
TMF720	Digital Identity	Mandatory	- GET - GET/id - POST - PATCH/id - DELETE/id
TMF688	Event Management	Optional	event - GET - GET/id
TMF701	Process Flow Management	Optional	processFlow GET

API ID	API Name	Mandatory / Optional	Operations
			• GET/id • POST • DELETE/id taskFlow: • GET • GET/id • PATCH/id

3.2. Dependent APIs

Following diagram illustrates API/Resource/Operation:

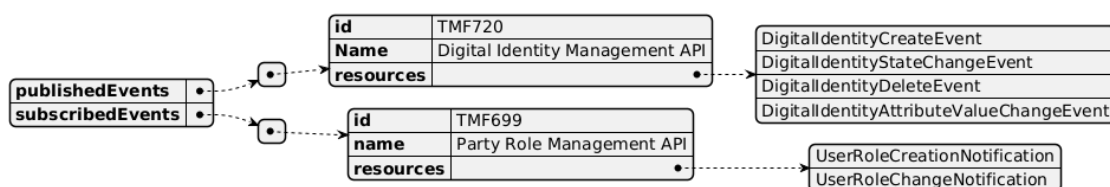


API ID	API Name	Mandatory / Optional	Operation
TMF632	Party Management	Conditional Mandatory (if Party Role is not provided)	individual • GET • GET/id organization • GET • GET/id
TMF669	Party Role Management	Conditional Mandatory (if Party Management is not provided)	partyRole • POST

API ID	API Name	Mandatory / Optional	Operation
TMF701	Process Flow Management	Optional	processFlow • POST • GET • GET/id • DELETE taskFlow • PATCH • GET • GET/id
TMF688	Event Management	Optional	event • GET • GET/id

3.3. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component table for the machine-readable component specification file for this component.

5. References

Refer to the ODA Component table for the machine-readable component specification file for this component.

5.1. TMF Standards related versions

Standard	Version(s)
SID	23.5
eTOM	23.5
Functional Framework	23.5

5.2. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	28 Jun 2022	Gaetano Biancardi , Sylvie Demarest	Final edits prior to publication
1.1.0		Amaia White	Final edits prior to publication
1.0.1	25 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14 Aug 2023	Amaia White	Final edits prior to publication
1.1.0	13 May 2024	Gaetano Biancardi	Component Template Update Component refreshed to ODF 23.5 (updates required only for functional framework functions) Removed TMF688 from exposed and dependent API

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	05-Aug-2022	Goutham Babu	Initial release of document
Pre-production	05-Sep-2022	Adrienne Walcott	Updated to reflect TM Forum Member Evaluated status
Pre-production	14-Aug-2023	Amaia white	New release of document 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-Production	12-Jul-2024	Amaia White	New release of document 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team:

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